AWS DEVOPS CLOUD COMPUTING INTERNSHIP

Hosting a Static Website using Amazon S3 and Amazon CloudFront

Project Report — Step-by-Step Implementation

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Mentors: Mrs Bethala Sumana • Mr Juttuka Lovababu

Team & Mentors

1

Sayyed Althamas

2

Sai Pradeep

3

CH Sivani

4

K Sofiya

Mentors

Mrs Bethala Sumana

Mr Juttuka Lovababu

OBJECTIVE

What We Set Out To Build

To design, deploy, and deliver a static portfolio-style website using core AWS cloud services — demonstrating practical, hands-on skills in cloud storage, global content delivery, and secure access control.

1. **Amazon S3**

   Object storage and static website hosting

2. **Amazon CloudFront**

   Global content delivery network (CDN)

3. **Origin Access Control**

   Secure, private access from CloudFront to S3

Tools & Services Used

1

Amazon S3

Store and host the static website files

2

Amazon CloudFront

CDN serving the website globally with low latency

3

Origin Access Control

Restrict S3 bucket access to CloudFront only

4

AWS Management Console

Used to configure all services

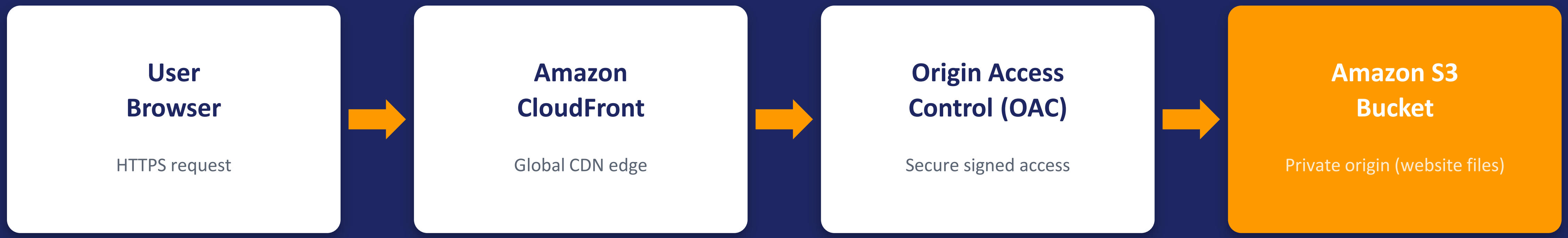
5

HTML & CSS

Build the website content (index.html, style.css)

Architecture Overview

How a visitor's request securely reaches the website



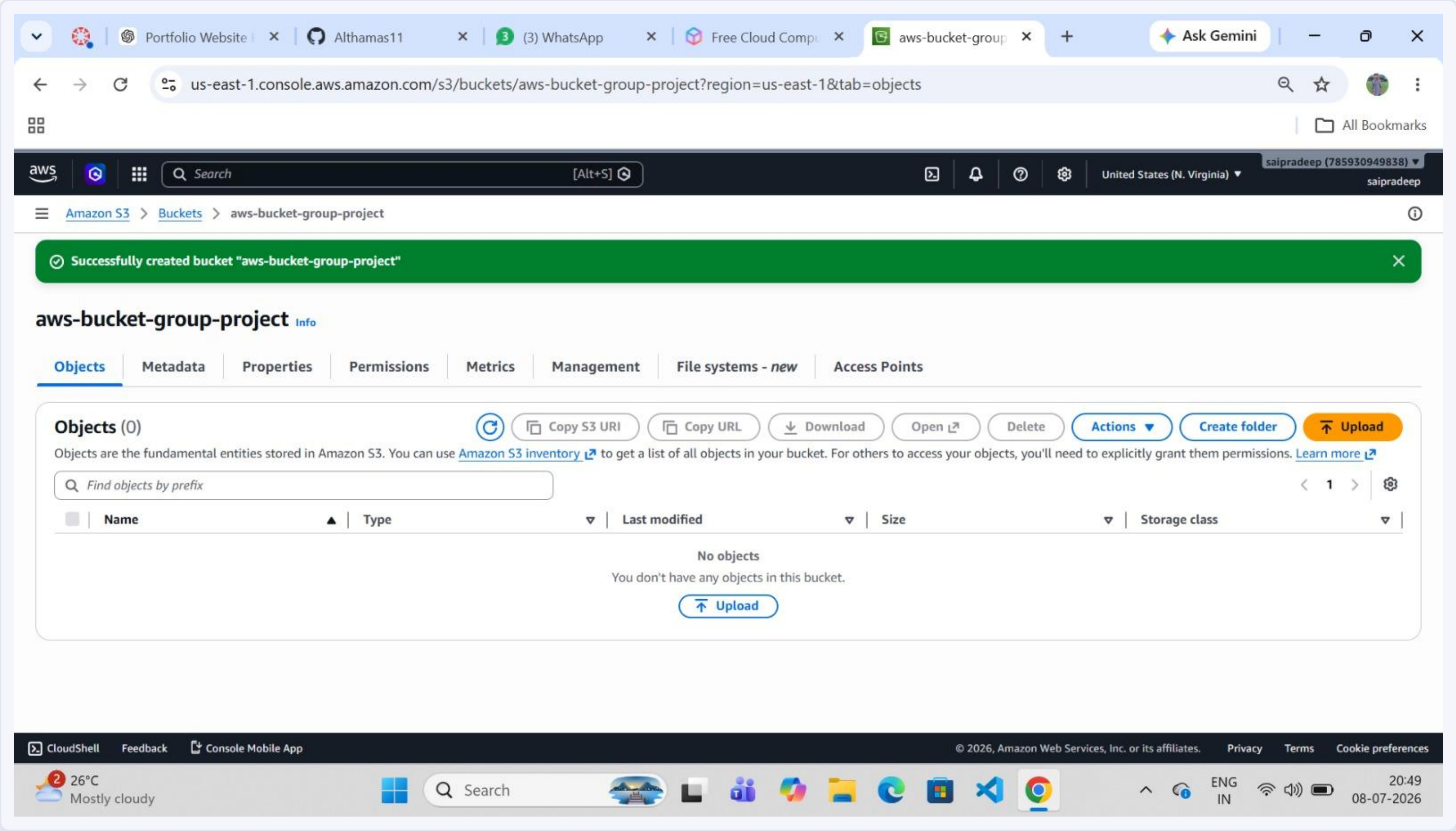
The S3 bucket stays fully private — Block Public Access is ON. Content is served to the world only through CloudFront, authenticated via OAC.

PHASE 1 · S3 SETUP

STEP

01

Create an Amazon S3 Bucket

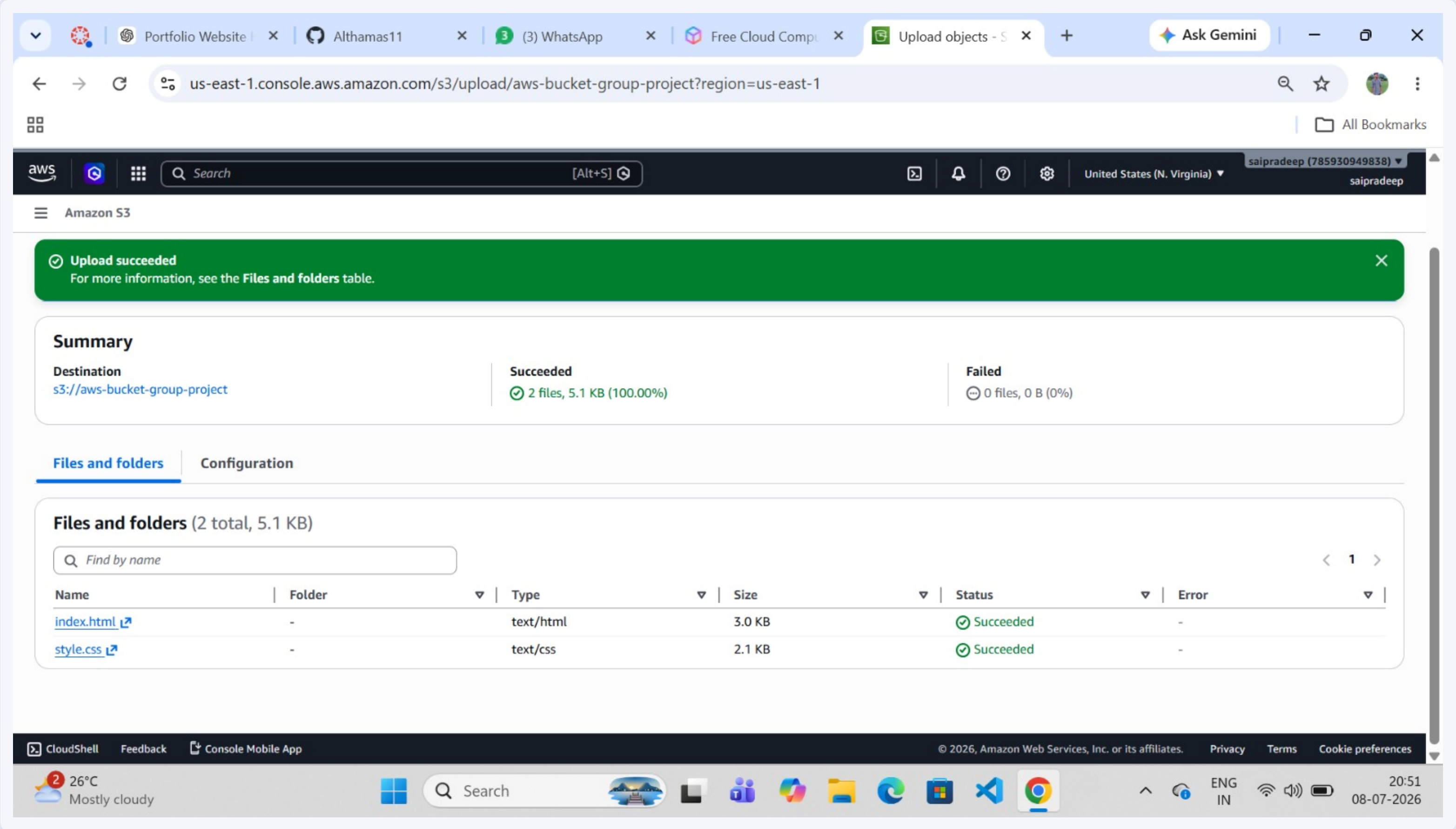


An S3 bucket named "aws-bucket-group-project" was created in the US East (N. Virginia) region to store the website's files.

STEP

02

Upload the Website Files



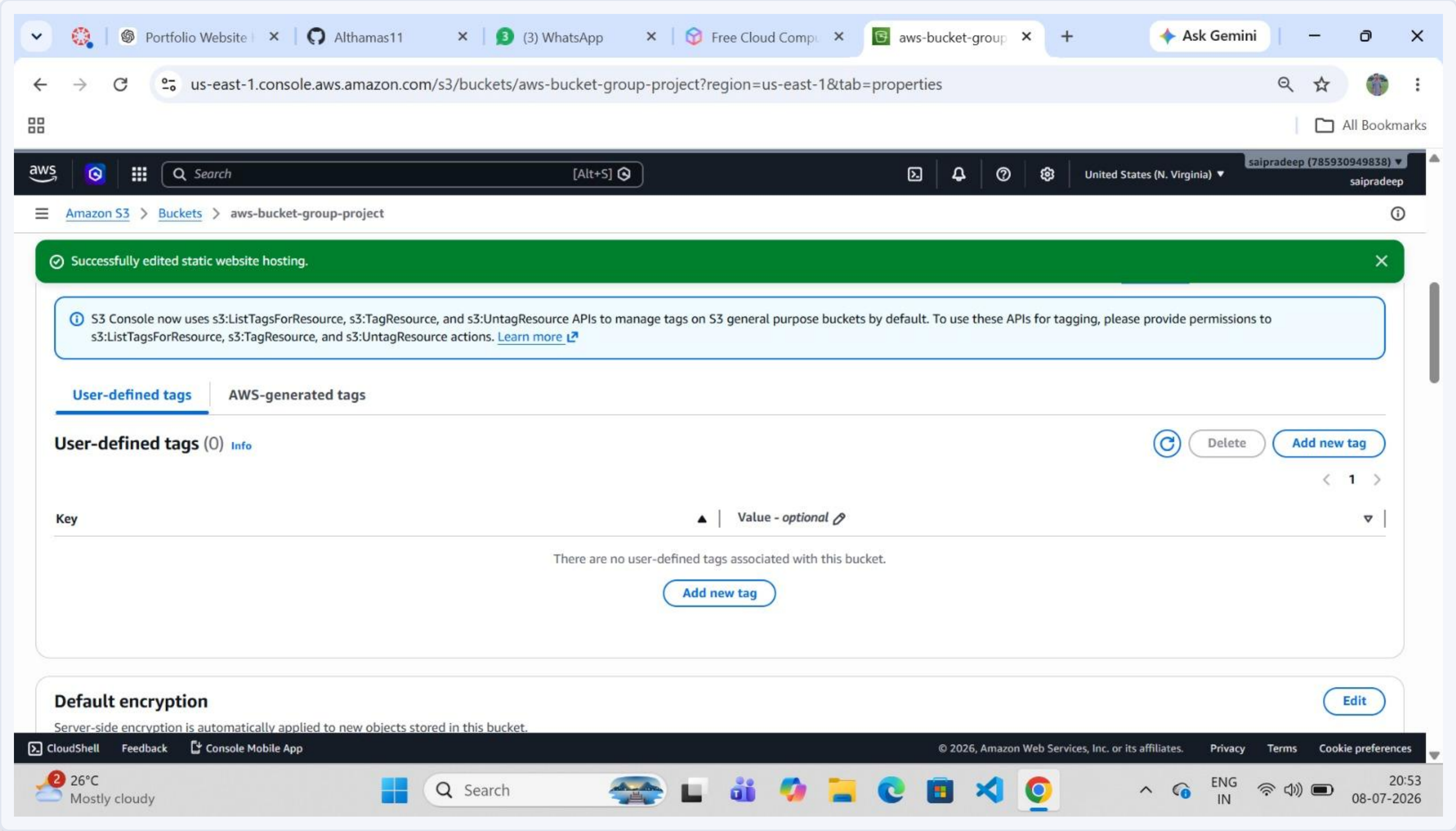
index.html and style.css were uploaded to the bucket. Upload summary confirms 2 files (5.1 KB total) succeeded.

PHASE 1 · S3 SETUP

STEP

03

Enable Static Website Hosting



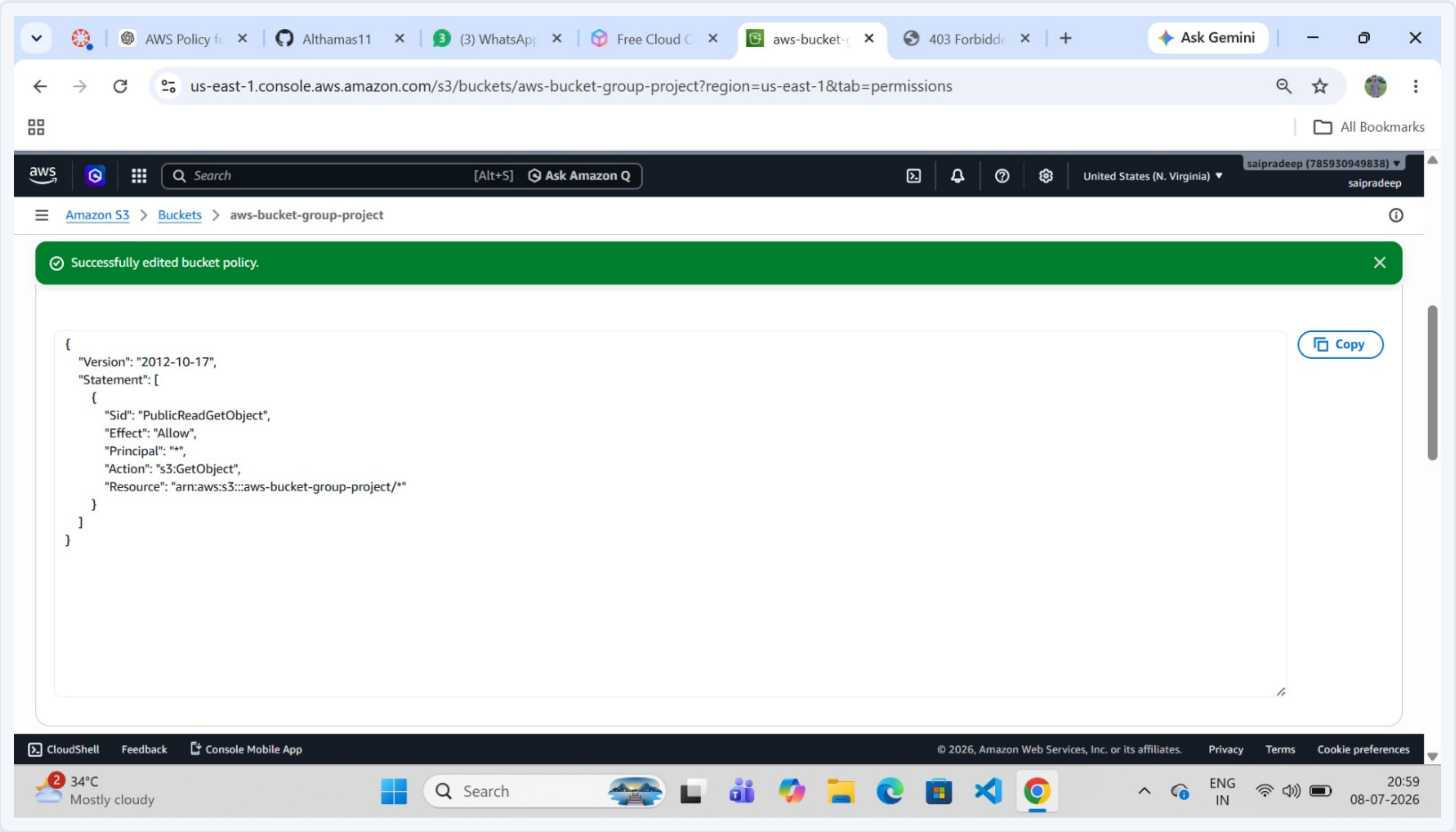
Static Website Hosting was enabled in the bucket's Properties tab, with index.html set as the index document.

PHASE 1 · S3 SETUP

STEP

04

Configure Initial Public-Read Policy



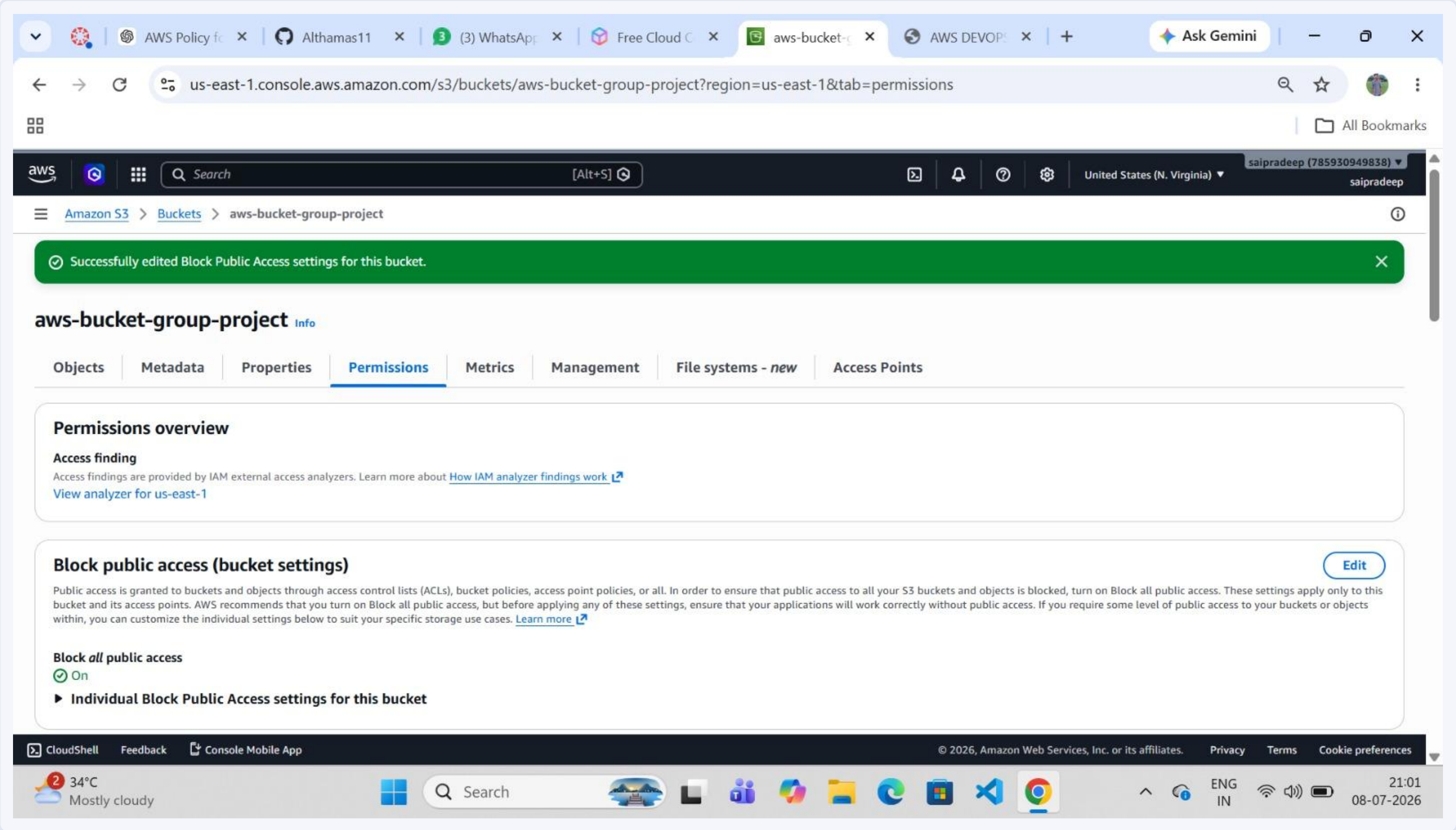
A PublicReadGetObject bucket policy was attached to allow public GetObject access — the traditional S3 hosting method.

PHASE 1 · S3 SETUP

STEP

05

Review Block Public Access Settings



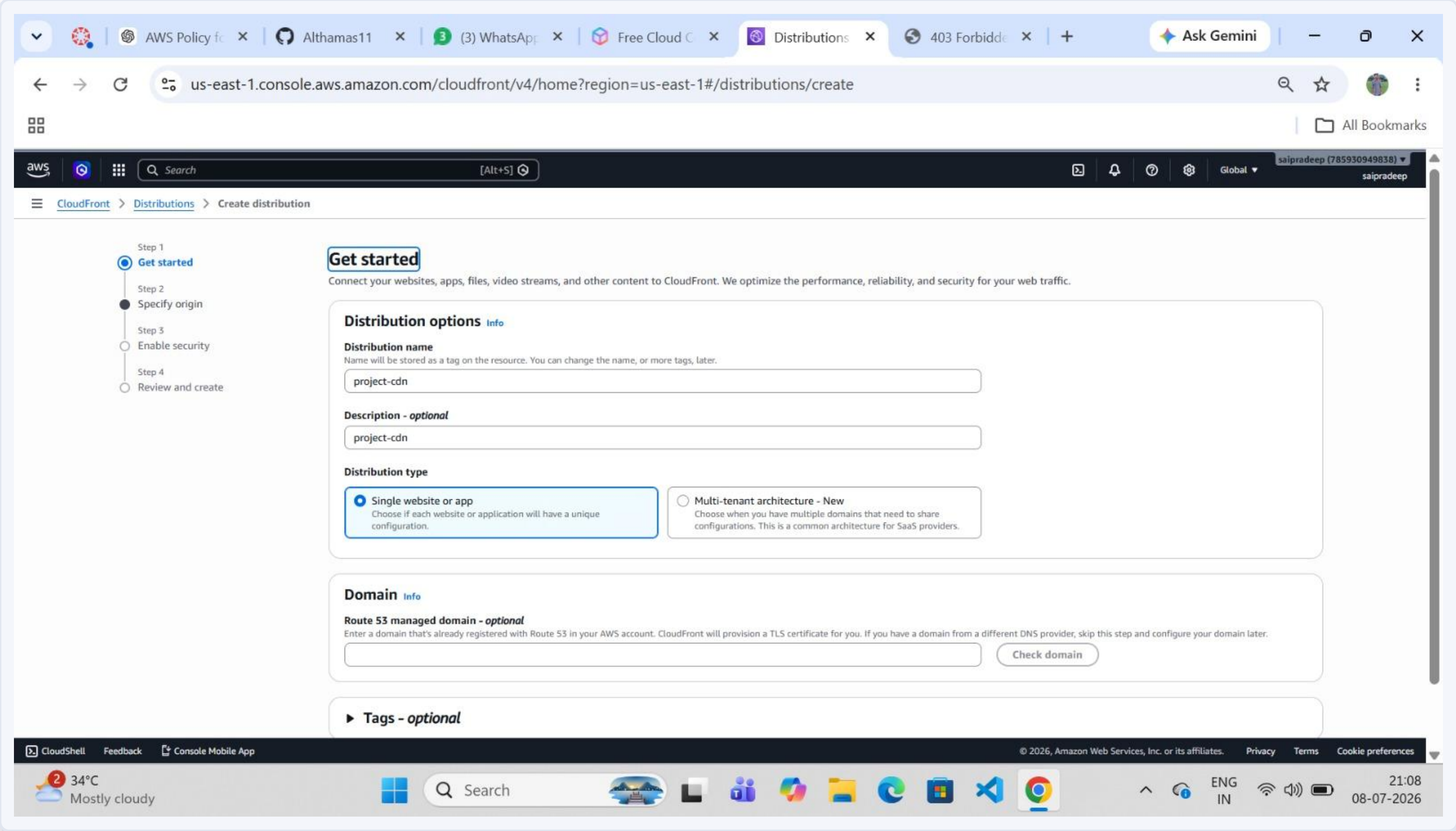
Block all public access was kept ON at the bucket level, following AWS security best practices — content is instead served via CloudFront.

PHASE 2 · CLOUDFRONT DISTRIBUTION

STEP

06

Create a CloudFront Distribution



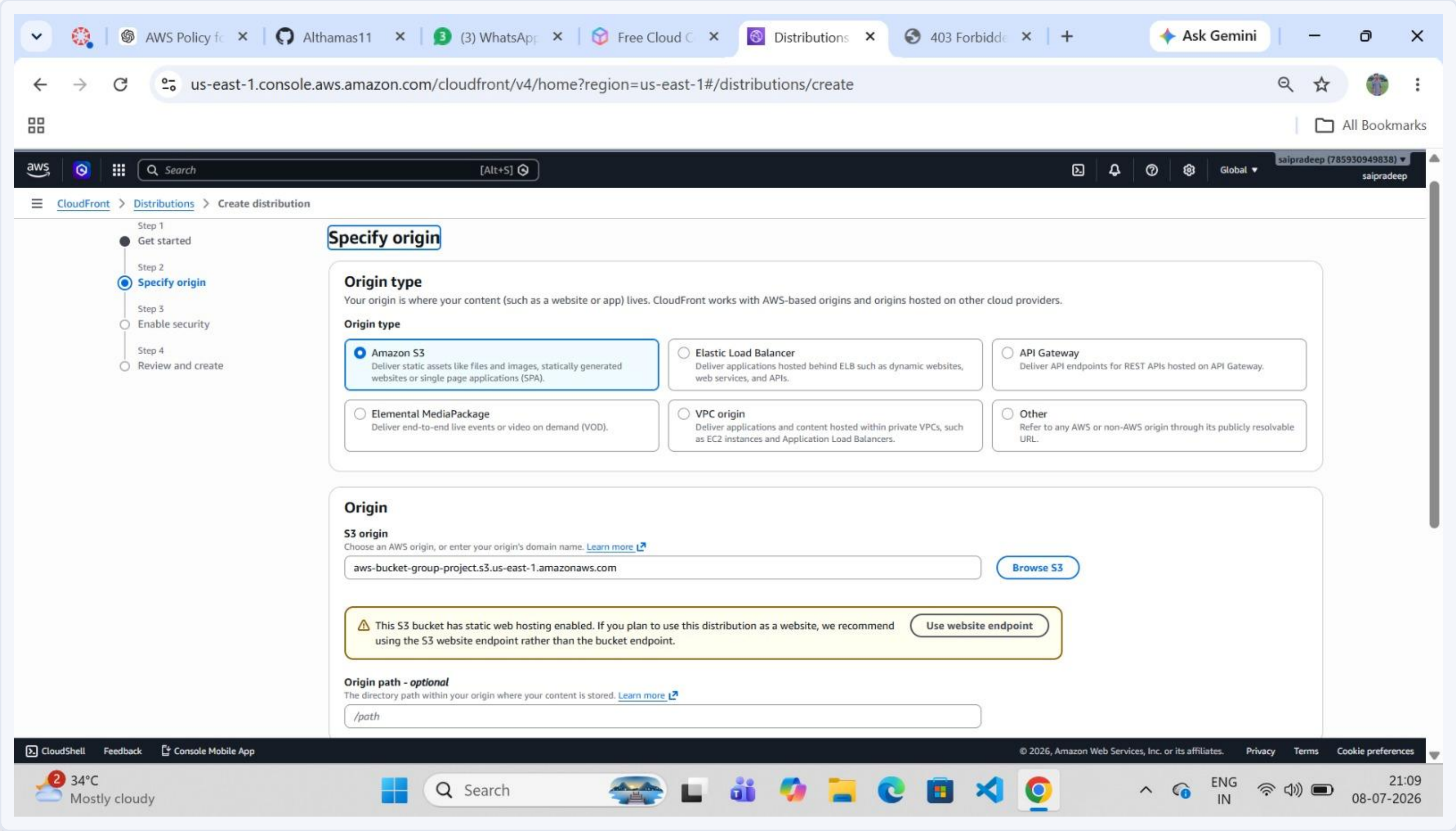
A new CloudFront distribution named "project-cdn" was created using the "Single website or app" distribution type.

PHASE 2 · CLOUDFRONT DISTRIBUTION

STEP

07

Specify the Origin (S3 Bucket)



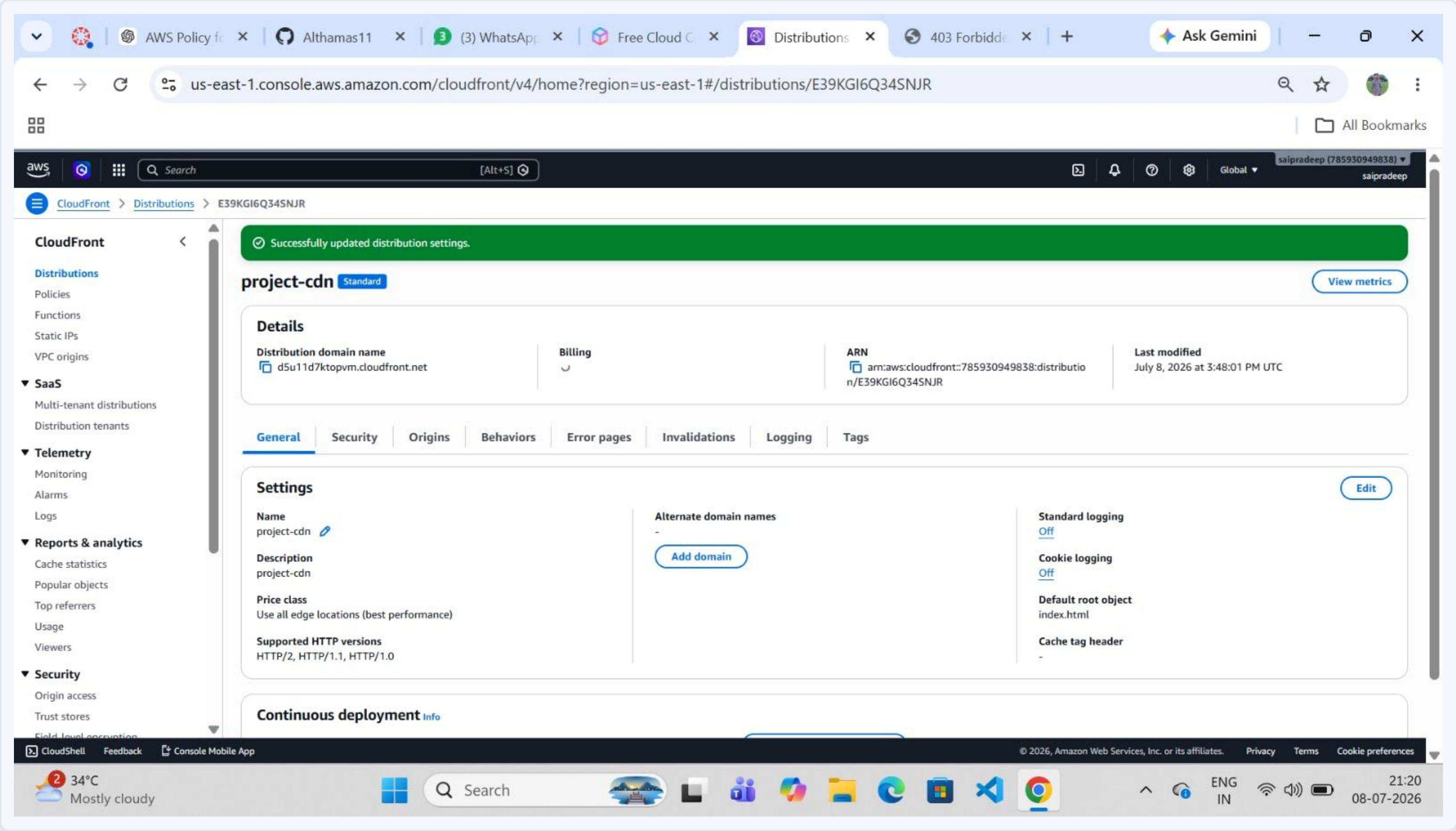
The origin type was set to Amazon S3, selecting aws-bucket-group-project.s3.us-east-1.amazonaws.com as the origin.

PHASE 2 · CLOUDFRONT DISTRIBUTION

STEP

08

Configure Origin Access Settings



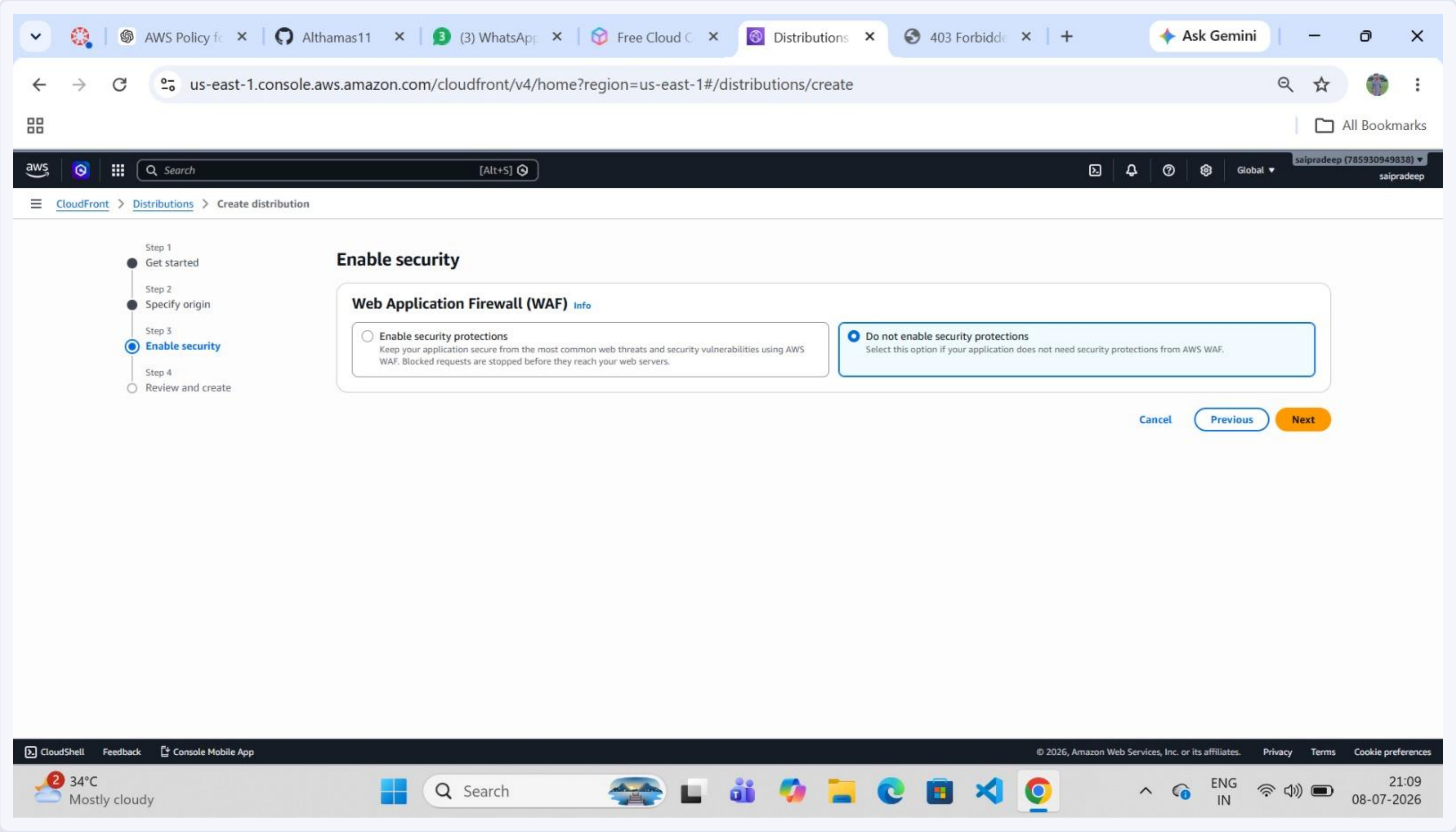
"Allow private S3 bucket access to CloudFront" (recommended) was enabled with recommended origin and cache settings.

STEP

09

Enable Security Settings (WAF)

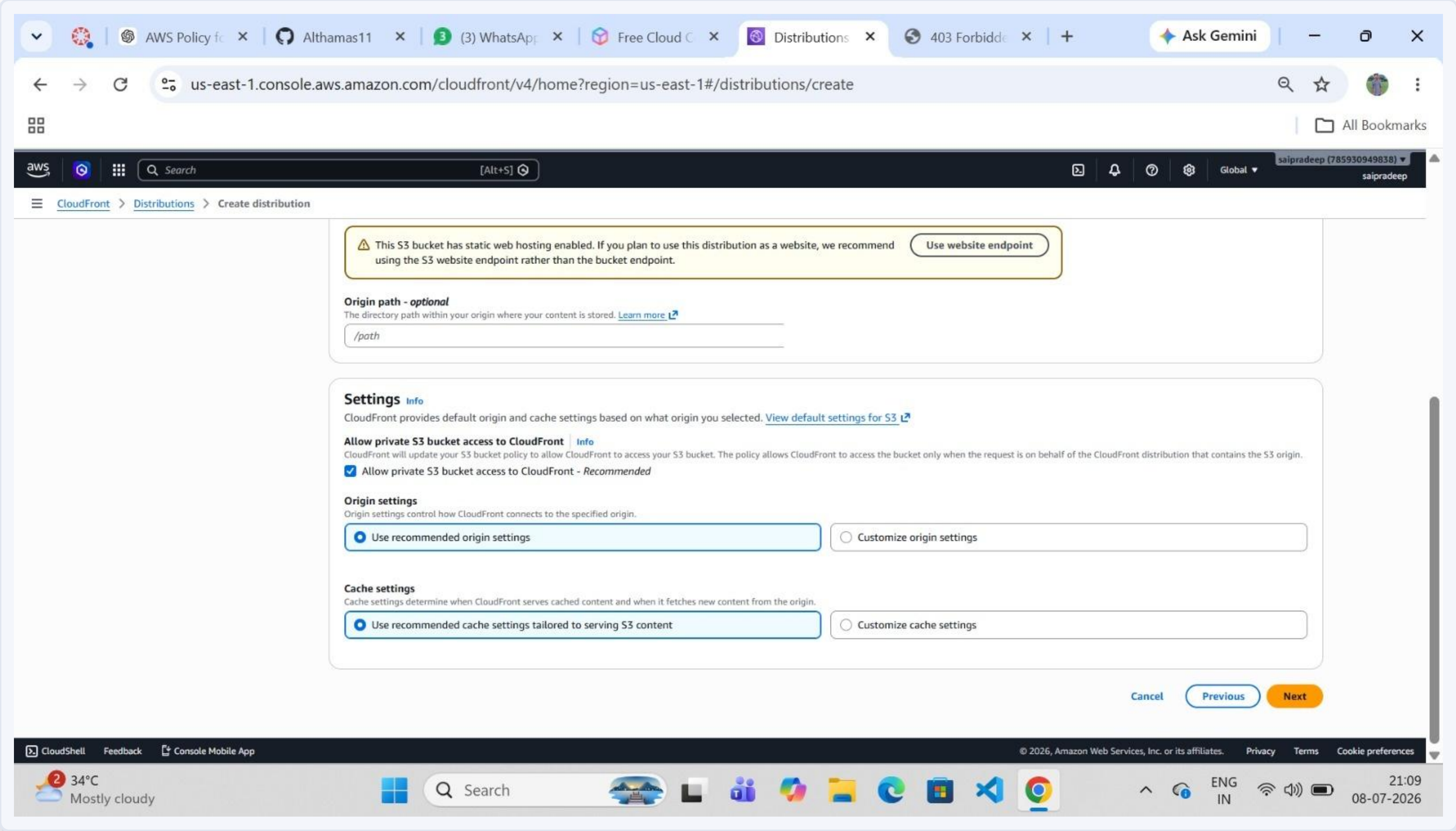
"Do not enable security protections" was selected, since AWS WAF was not required for this internship-scale project.



STEP

10

Review & Create the Distribution

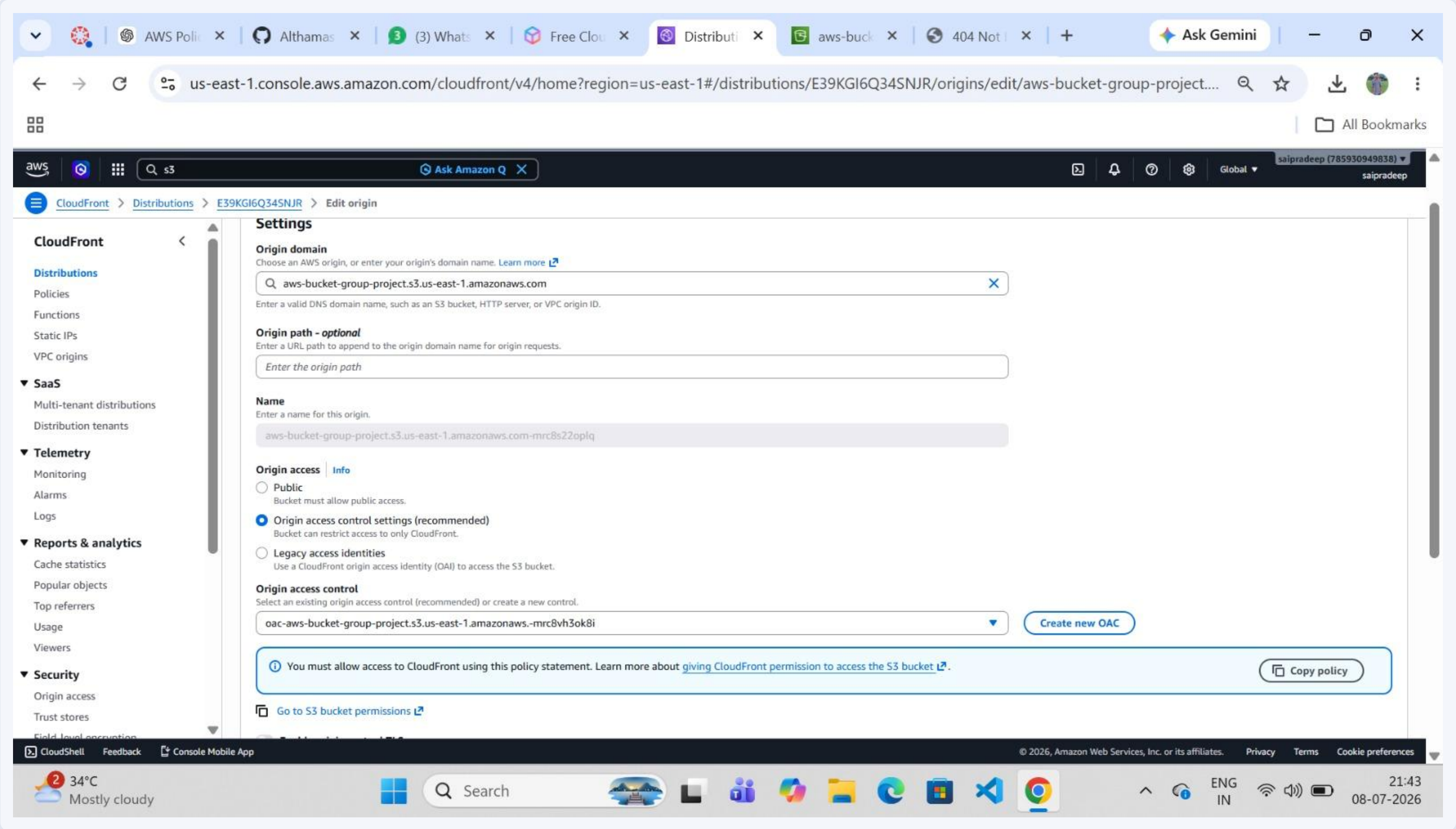


The distribution "project-cdn" was created successfully and assigned the domain d5u11d7ktopvm.cloudfront.net.

STEP

11

Configure Origin Access Control (OAC)



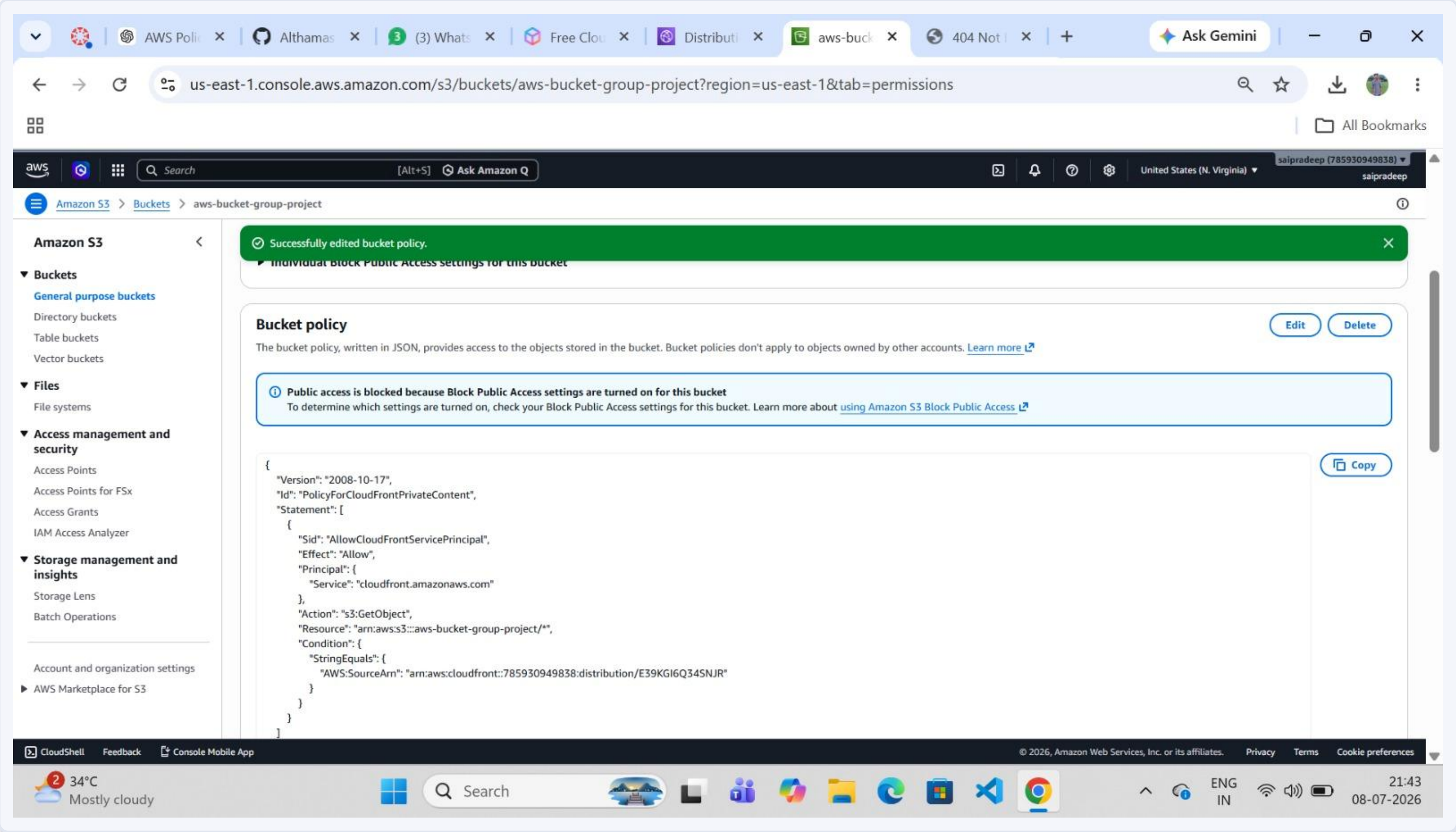
The origin was edited to use "Origin access control settings (recommended)", and a new OAC was created and attached.

STEP

12

Bucket Policy Updated for CloudFront

CloudFront auto-generated an updated bucket policy (AllowCloudFrontServicePrincipal), replacing the earlier public-read policy.



STEP

13

Access the Website via CloudFront

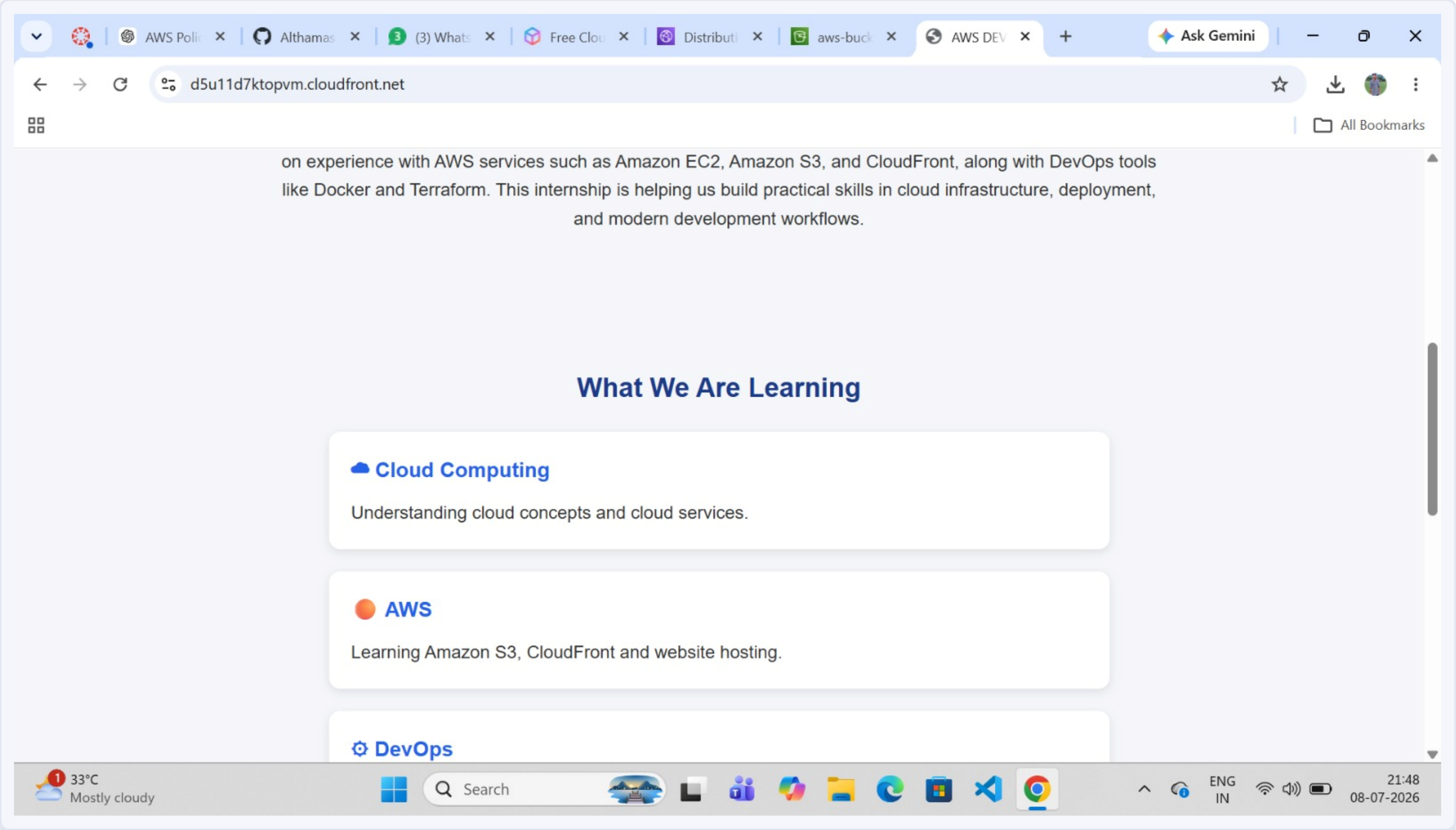
The website was accessed via d5u11d7ktopvm.cloudfront.net, confirming the homepage banner, navigation, and welcome section load.

PHASE 4 · LIVE VERIFICATION

STEP

14

Verify the About / Learning Section



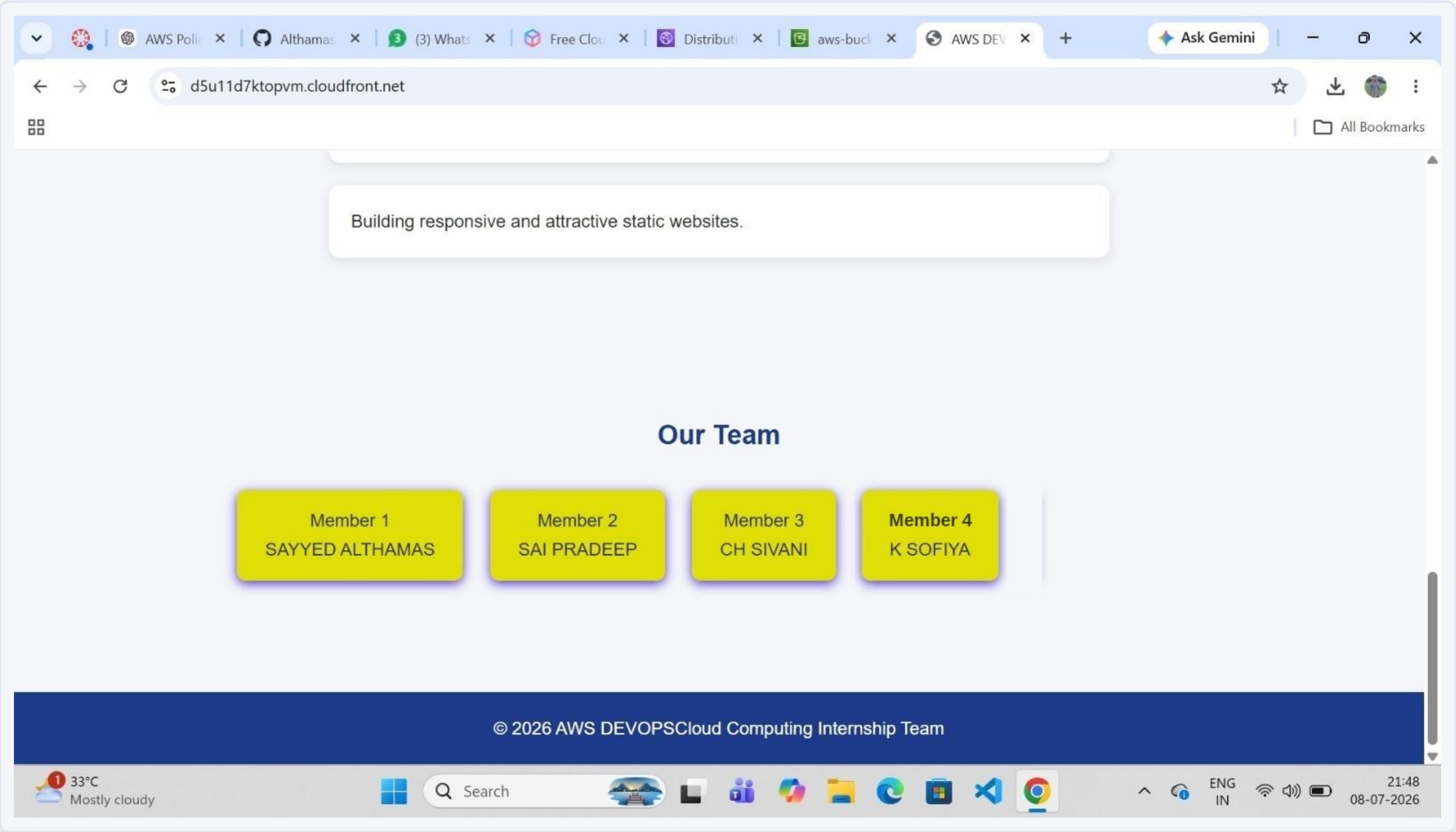
The "About Our Internship" and "What We Are Learning" sections render correctly, listing Cloud Computing, AWS, and DevOps.

PHASE 4 · LIVE VERIFICATION

STEP

15

Verify the Team Section



The "Our Team" section displays all team members along with the footer copyright — confirming the site is fully live via CloudFront.

Conclusion

This project demonstrates end-to-end static website deployment on AWS: storing files in Amazon S3, enabling static website hosting, and securely distributing content worldwide using Amazon CloudFront with Origin Access Control.

The S3 bucket stays private while the website remains publicly and securely accessible via CloudFront's global edge network — reflecting real-world AWS DevOps best practices.

Amazon S3 | Amazon CloudFront | Origin Access Control | Static Hosting | Global CDN

Thank You